

GATE 2009, ECE Question Number 39

Abstract

This project implements the JK Flip-Flop-based 3-state counter problem from GATE 2009 Q39 using Raspberry Pi Pico W. Through simulation of JK flip-flop behavior, button-triggered clock inputs change the output state, which is observed on two LEDs representing Q1 and Q2. This project demonstrates how sequential logic circuits can be modeled using embedded hardware and MicroPython programming.

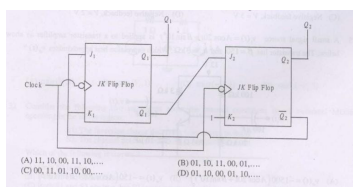


Figure: JK Flip-Flop logic diagram used in simulation

1. Components

Component	Qty
Raspberry Pi Pico W	1
Push Button (clock input)	1
LEDs (Q1, Q2)	2
220 Ω Resistors	2
Breadboard	1
Jumper Wires	8
Micro-USB Cable	1

Table 1: List of components used

2. Setup and Connections

- **Button (Clock)** $\rightarrow$ GPIO 15 with a pull-down resistor.
- **LED Q1** $\rightarrow$ GPIO 16 via 220 Ω resistor.
- **LED Q2** $\rightarrow$ GPIO 17 via 220 Ω resistor.
- GND connections from Pico to breadboard.
- Button press triggers state change.

3. State Table for Q39

Press	Q2	Q1
Initial	0	0
1st	0	1
2nd	1	0
3rd	0	0
4th	0	1

Table 2: State transition sequence for 3-state counter

4. Analysis

- Flip-Flop 1 toggles on every clock ($J=Q_2$, $K=1$)
- Flip-Flop 2 toggles based on Q_1 ($J=Q_1$, $K=1$)
- Implemented logic replicates JK Flip-Flop transitions
- LED outputs match the cyclic state sequence: $01 \rightarrow 10 \rightarrow 00 \rightarrow \dots$

5. Circuit Image

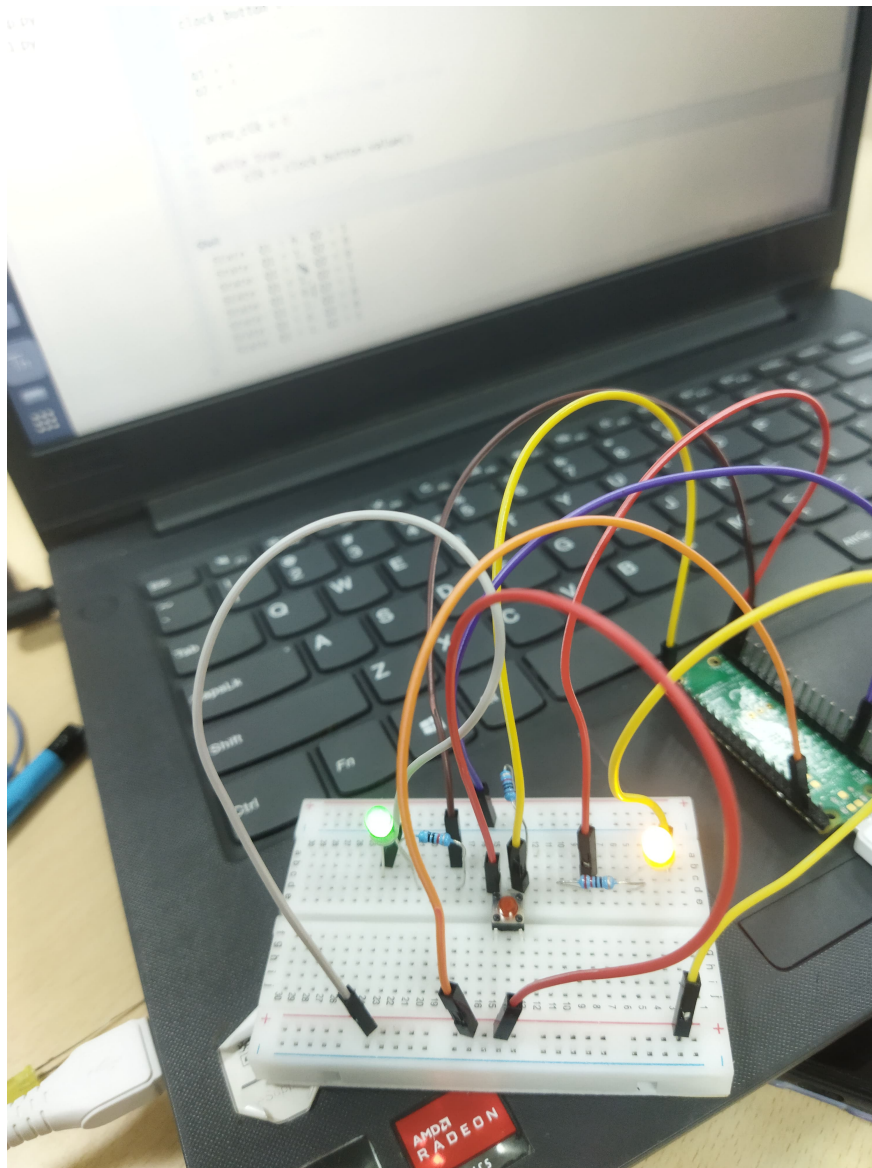


Figure: Real hardware setup using Raspberry Pi Pico W

6. Conclusion

This hardware implementation using Raspberry Pi Pico W accurately simulates the JK flip-flop logic as described in GATE ECE 2009 Q39. It clearly demonstrates the principles of sequential logic and state transitions through LED outputs. This experiment helps bridge theoretical flip-flop concepts with practical embedded systems. The modularity of Pico W also makes the solution scalable for more complex digital systems.

7. GitHub Repository

- Project code and documentation:
https://github.com/Rithrishareddy/RITHRISHA_FWC/tree/main/Hardware

8. Learning Outcomes

- Gained hands-on experience with JK flip-flop implementation on real hardware.
- Understood sequential circuit behavior through state transitions.
- Practiced GPIO pin configuration, debouncing techniques, and LED state control.
- Learned to simulate theoretical counters in a hardware-based environment.
- Improved MicroPython programming and debugging skills on embedded platforms.